+91 82521 94251
Bengluru, India
roshankumar951@gmail.com

Roshan Kumar

Software Engineer | Android Developer

Portfolio: Portfolio
github.com/RoshanKumar404
LinkedIn:LinkedIn

B.Tech CSE graduate (2026) who builds Android and React Native apps, mostly in Kotlin, Jetpack Compose and TypeScript, with Node.js or Python backends on PostgreSQL. Currently building an offline-first inventory app with its own sync engine. Solid on DSA, OS, DBMS and computer networks.

Skills

Languages	TypeScript, JavaScript, Kotlin, Java, Python, SQL
Mobile	React Native, Android SDK, Jetpack Compose, MVVM, Kotlin Coroutines, Android Background Listeners
Backend & Databases	Node.js, Flask, REST APIs, JWT Authentication, PostgreSQL, PostGIS, pgvector, MongoDB, Firebase
ML & Tools	TensorFlow Lite, MobileNetV2, Git, GitHub, Docker, Linux, Android Studio, VS Code, Cloudinary
Core CS	Data Structures, Algorithms, OOP, Operating Systems, DBMS, Computer Networks, System Design

Technical Experience

Android Developer Intern, Apex Planet **May 2025 – June 2025**
Kotlin, MVVM, Coroutines, REST APIs, Performance Profiling

- Connected 2 **REST** endpoints using **Coroutines**, so data loads in the background and the app doesn't crash when the network drops.
- Fixed laggy list scrolling (30 to 40 FPS) by moving data processing off the main thread and trimming list rendering. Found the problem with the profiler.

Projects

Kirana Ledger: Offline-First Inventory & Credit App **In Progress**
React Native, TypeScript, SQLite, Node.js, PostgreSQL

- Building an offline-first inventory, billing and **udhaar** (customer credit) app for small kirana shops in **React Native**, with **SQLite** on the phone so a shop can keep selling when the network drops.
- Storing stock and customer balances as **append-only ledger entries** instead of one editable number, so two phones selling offline don't overwrite each other.
- Writing the **sync layer** from scratch on a **Node.js** and **PostgreSQL** backend: changes go into a queue and are retried safely, in batches of 50 per upload with the retry delay capped at 5 minutes.

CryptNote: Personal Knowledge Vault **V1.0.1**
React Native,Client-Side Encryption, PostgreSQL, pgvector, AI Embeddings, Semantic Search

- Built a private vault for notes, ideas, credential details and code snippets. Each note is **encrypted on the client** with AES-256-GCM before it is sent to the backend.
- Added **semantic search**: notes get embeddings and stored in **PostgreSQL** with **pgvector**.
- Search the note with a single word or whole sentence across notes.
- A plain-language query returns the top 5 matching notes.

Personal Finance Advisor (Hackathon Finalist) **Sept 2025**
React Native, Android Background Listeners, On-Device Storage

- Built a **React Native** expense tracker with Android **background listeners** that pick up bank and UPI notifications and turn 1 message formats into transactions, with no manual entry.
- Transactions and balances stay on the phone, so it works **fully offline** and no financial data leaves the device.
- Spending is sorted into 3 categories using simple rules, and the app suggests budgeting tips based on what is left for the month.

Education

Bachelor of Technology in Computer Science and Engineering *Ramgarh Engineering College (Jharkhand University of Technology)*
2022 – 2026

Relevant Coursework: Data Structures, Algorithms, Operating Systems, DBMS, Computer Networks, Object-Oriented Programming, System Design

Higher Secondary (PCM)
Guru Gobind Singh Public School, Ranchi
2019 – 2021